

Basic – Calculator Program

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Introduction

> This project is a simple calculator program developed using the C programming language. The program performs basic arithmetic operations such as addition, subtraction, multiplication, and division using switch-case statements. It helps beginners understand conditional statements and user input handling in C.

Objective

- To perform arithmetic operations in C
- To understand switch-case statements
- To learn user input and output handling

Tools & Technologies Used

Tool	Purpose
C Programming	Program development
GCC Compiler	Compilation

Source Code

```
#include <stdio.h>

int main() {
    float num1, num2, result;
    int choice;

    // Input two numbers
    printf("Enter first number: ");
    scanf("%f", &num1);

    printf("Enter second number: ");
    scanf("%f", &num2);

    // Display menu
    printf("\nSelect Operation:\n");
    printf("1. Addition\n");
    printf("2. Subtraction\n");
    printf("3. Multiplication\n");
    printf("4. Division\n");
```

```
printf("Enter your choice (1-4): ");
```

```
scanf("%d", &choice);
```

```
// Perform operation using switch case
```

```
switch(choice) {
```

```
    case 1:
```

```
        result = num1 + num2;
```

```
        printf("Result = %.2f\n", result);
```

```
        break;
```

```
    case 2:
```

```
        result = num1 - num2;
```

```
        printf("Result = %.2f\n", result);
```

```
        break;
```

```
    case 3:
```

```
        result = num1 * num2;
```

```
        printf("Result = %.2f\n", result);
```

```
        break;
```

```
    case 4:
```

```
        if(num2 != 0) {
```

```
            result = num1 / num2;
```

```
            printf("Result = %.2f\n", result);
```

```
        } else {
```

```
            printf("Error! Division by zero is not allowed.\n");
```

```
        }
```

```
        break;
```

```
        default:
            printf("Invalid choice!\n");
    }

    return 0;
}
```

Explanation of Code

- The program uses `stdio.h` for input and output functions.
- `main()` is the starting function of the program.
- Variables are declared to store numbers, result, and user choice.
- `scanf()` is used to take two numbers as input from the user.
- A menu is displayed to select an arithmetic operation.

Switch statement performs:

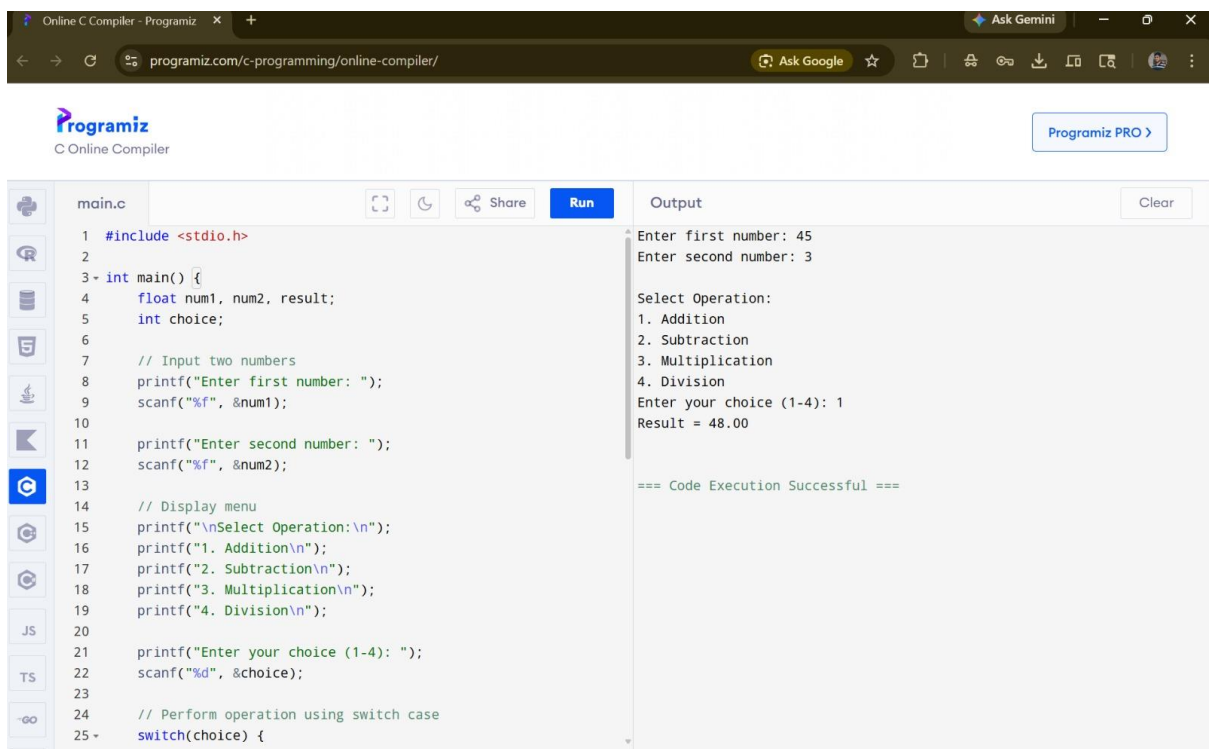
- Addition
- Subtraction
- Multiplication
- Division

The program checks division by zero before dividing.

default case handles invalid choices.

return 0 ends the program successfully.

Output



The screenshot shows the Programiz Online C Compiler interface. The code editor on the left contains a C program for a calculator. The output window on the right shows the program's execution results.

```
main.c
1 #include <stdio.h>
2
3 int main() {
4     float num1, num2, result;
5     int choice;
6
7     // Input two numbers
8     printf("Enter first number: ");
9     scanf("%f", &num1);
10
11    printf("Enter second number: ");
12    scanf("%f", &num2);
13
14    // Display menu
15    printf("\nSelect Operation:\n");
16    printf("1. Addition\n");
17    printf("2. Subtraction\n");
18    printf("3. Multiplication\n");
19    printf("4. Division\n");
20
21    printf("Enter your choice (1-4): ");
22    scanf("%d", &choice);
23
24    // Perform operation using switch case
25    switch(choice) {
```

Output

```
Enter first number: 45
Enter second number: 3

Select Operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 1
Result = 48.00

=== Code Execution Successful ===
```

Online C Compiler - Programiz

programiz.com/c-programming/online-compiler/

Programiz C Online Compiler

main.c

```
25 switch(choice) {
26     case 1:
27         result = num1 + num2;
28         printf("Result = %.2f\n", result);
29         break;
30
31     case 2:
32         result = num1 - num2;
33         printf("Result = %.2f\n", result);
34         break;
35
36     case 3:
37         result = num1 * num2;
38         printf("Result = %.2f\n", result);
39         break;
40
41     case 4:
42         if(num2 != 0) {
43             result = num1 / num2;
44             printf("Result = %.2f\n", result);
45         } else {
46             printf("Error! Division by zero is not allowed\n");
47         }
48         break;
```

Output

Enter first number: 45
Enter second number: 3

Select Operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 1
Result = 48.00

=== Code Execution Successful ===

Online C Compiler - Programiz

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Programiz C Online Compiler

main.c

```
32     result = num1 - num2;
33     printf("Result = %.2f\n", result);
34     break;
35
36     case 3:
37         result = num1 * num2;
38         printf("Result = %.2f\n", result);
39         break;
40
41     case 4:
42         if(num2 != 0) {
43             result = num1 / num2;
44             printf("Result = %.2f\n", result);
45         } else {
46             printf("Error! Division by zero is not allowed\n");
47         }
48         break;
49
50     default:
51         printf("Invalid choice!\n");
52 }
53
54 return 0;
55 }
```

Output

Enter first number: 45
Enter second number: 3

Select Operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 1
Result = 48.00

=== Code Execution Successful ===

Code Execution Screenshots

Output

Clear

```
Enter first number: 45
Enter second number: 3

Select Operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 1
Result = 48.00

=== Code Execution Successful ===
```

Screenshot of Performing Addition Operation

Output

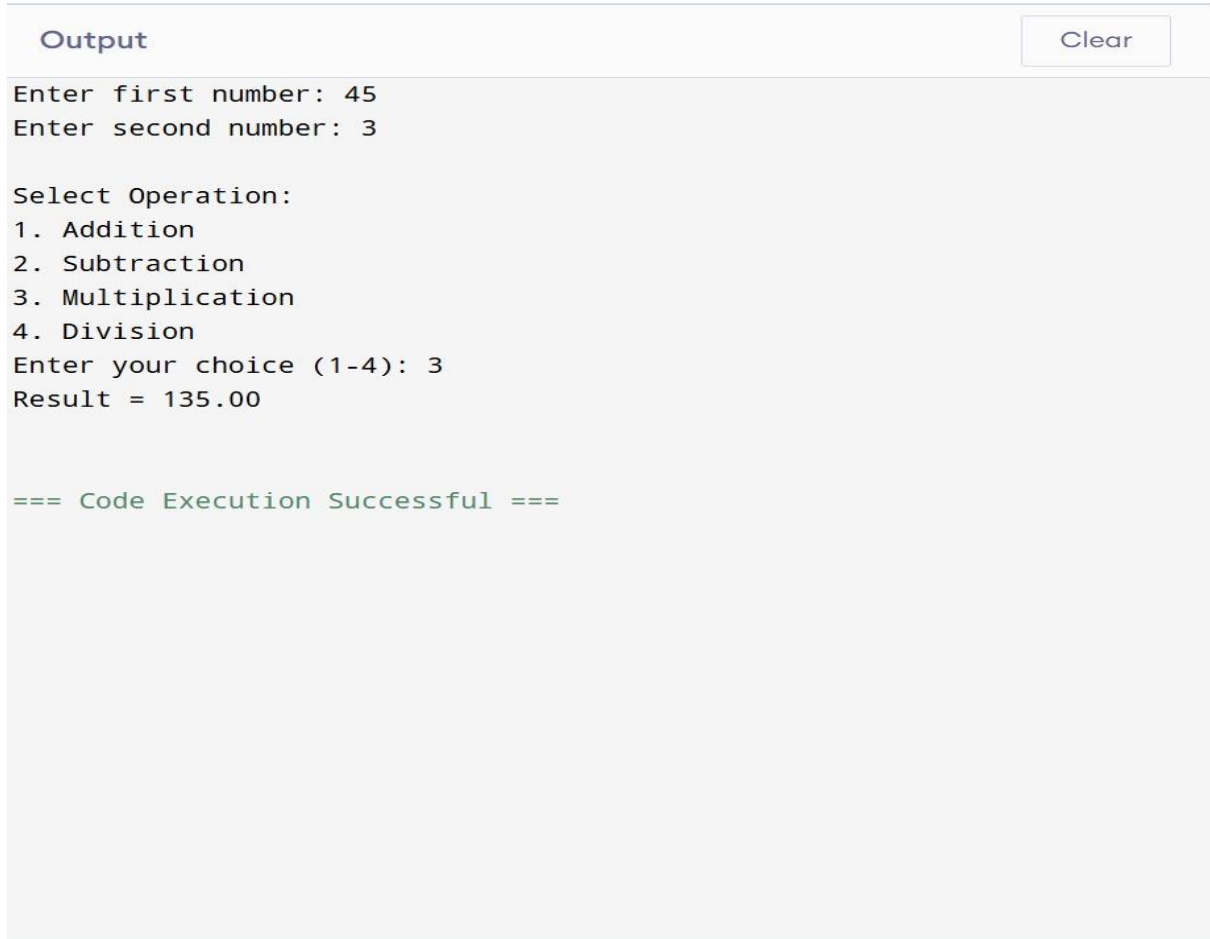
Clear

```
Enter first number: 45
Enter second number: 3

Select Operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 2
Result = 42.00

=== Code Execution Successful ===
```

Screenshot of Performing Subtraction Operation



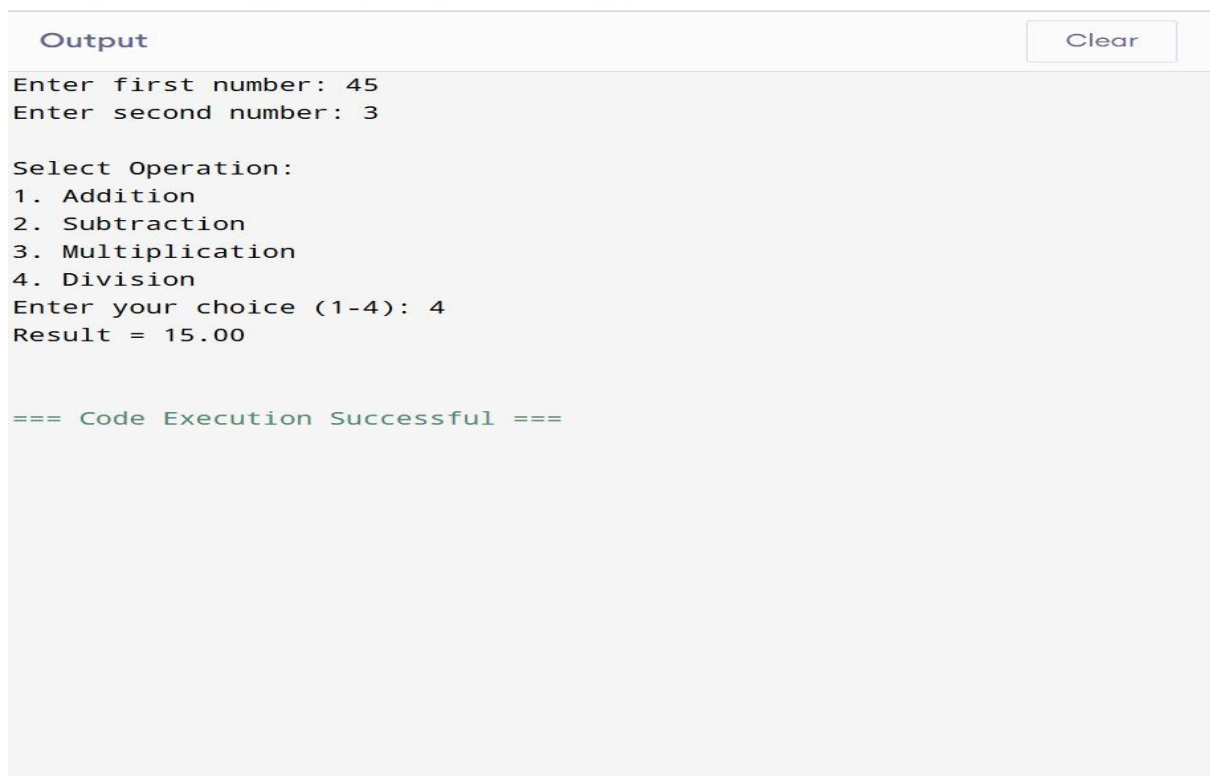
The screenshot shows a code execution environment with a header bar labeled 'Output' and a 'Clear' button. The output text is as follows:

```
Enter first number: 45
Enter second number: 3

Select Operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 3
Result = 135.00

=== Code Execution Successful ===
```

Screenshot of Performing Multiplication Operation



The screenshot shows a terminal window titled "Output" with a "Clear" button in the top right corner. The terminal displays the following text:

```
Enter first number: 45
Enter second number: 3

Select Operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 4
Result = 15.00

=== Code Execution Successful ===
```

Screenshot of Performing Division Operation

Conclusion

- This project successfully performs basic arithmetic operations such as addition, subtraction, multiplication, and division using the C programming language. It helped in understanding important concepts like user input, switch-case statements, and arithmetic operators in C.